Bridging functional and network robustness

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Table of contents

The idea

The role of a species in a community can be defined in three ways: where the species is in the functional space, where the species is in the ecological network, and what is the impact of the species loss. These are three lenses with which we can look at a species. They are usually used separately. Yet, they all are an attempt to capture the very same reality. So we argue that to improve our view on a species role in a community it is best to pile up these lenses than using them separately. They are complementary facets and understanding the relationships between these different view enable a more complete view of a role played by a species within a community.

The question

To what degree do a species position in functional trait space, its structural position in the interaction network, and its contribution to community robustness converge or diverge? What does that tell us about community robustness?

The axes

1. From static correlation to mechanistic understanding

Coux et al. (2016) established that functional originality correlates with network specialisation in pollinators. But the why remains underexplored. Traits govern which interactions are possible, such as morphological matching between flower and pollinators. So the trait–network relationship is not merely correlative but mechanistic. A promising direction is to move beyond summary axes (PCoA) and ask which specific traits drive which network metrics. Disentangling these trait-by-metric relationships would give the correlation a mechanistic foundation.

2. Connecting functional position to extinction dynamics

A key question whether functional originality of the lost species determines the magnitude of that loss. If a species is functionally original and topologically central, does its removal trigger a disproportionate cascade? Or do these two dimensions predict different things?

3. Generality across systems

Coux et al. (2016) worked exclusively with plant-pollinator networks, an extension is to test whether the same framework holds in other interaction types — food webs, seed dispersal networks — where the trait–interaction link operates differently. If the relationship between functional position and network role is general, it points to a universal principle of community organisation. If it is system-specific, that contrast itself becomes informative about how different ecological architectures buffer or amplify the consequences of species loss.

Notes Ismaël

- Coux et al. (2016):
  - Code and data
  - **Originality**: distance to the community average
  - **Uniqueness**: distance to the nearest neighbour
  - PCoA is used to summarise traits, could investigate the role of each trait individually in explaining species position in the network.

Coux, Camille, Romina Rader, Ignasi Bartomeus, and Jason M. Tylianakis. 2016. "Linking Species Functional Roles to Their Network Roles." *Ecology Letters* 19 (7): 762–70. https://doi.org/10.1111/ele.12612.